

Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)

Rephrase the following statement in bi-conditional form
"If you get up early in the morning, you will be healthy"

Answer (Please select your correct option)

☐ You will be healthy if and only if you get up early in the morning

correct

☐ If you will be healthy then you will get up early in the morning

☐ If you will get up early in the morning then you will be healthy

☐ None of these

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Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)

Reductio ad absurdum law is symbolically denoted as

Answer (Please select your correct option)

☐ $(p \wedge q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$

☐ $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

☐ $p \rightarrow q \equiv \sim p \vee q$

☐ $p \rightarrow q \equiv (p \wedge \sim q) \rightarrow c$

correct

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Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

A critical row is that in which _____ premises have truth value T.

Answer (Please select your correct option)

☐ at least one

☐ exactly one

☐ all

☐ at least two

correct

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Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

What will be the output of an OR-gate if it has inputs 0 and 1?

Answer (Please select your correct option)

☐ 0

☐ 1

correct

☐ 2

☐ 3

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Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

Let U be the universal set and A is its subset then $A \cup A^c$ is equal to

Answer (Please select your correct option)

☐ A

correct

☐ A^c

☐ ϕ

☐ U

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Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

Identify the false statement

Answer (Please select your correct option)

☐ $0 \in \phi$

correct

☐ $\{\phi\} \subseteq \{\phi\}$

☐ If A and B are two sets $A \subseteq B$ and $B \subseteq A$ then $A = B$

☐ Two sets are disjoint if their intersection is empty set

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Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

If A and S are two reflexive relations then $A \cap S$ will be

Answer (Please select your correct option)

☐ Symmetric

☐ Reflexive

correct

☐ Transitive

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Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min)

Symmetric and Anti-symmetric relations are

Answer (Please select your correct option)

☐ negative of each other.

☐ same.

☐ not negative of each other.

correct

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Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

If two relations are reflexive then their composition is

Answer (Please select your correct option)

☐ Anti-symmetric

☐ Reflexive

correct

☐ Irreflexive

☐ Symmetric

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Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)

Inverse of relation can be obtained by

Answer (Please select your correct option)

☐ changing signs of elements in order pairs.

☒ changing position of elements in order pairs.

☐ taking multiplicative inverse of elements in order pairs.



correct

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Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min)

Let $A \times A = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$, find which one of the given relations is a function.

Answer (Please select your correct option)

☐

$$R_1 = \{(1,3), (2,2), (3,1)\}$$

☐

$$R_2 = \{(1,1), (1,2), (2,1)\}$$

☐

$$R_3 = \{(2,2), (2,3), (3,1)\}$$

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Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)

If $f(x) = 2x + 1$ and $g(x) = x^2 - 1$ then $f \circ f(x) =$

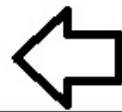
Answer (Please select your correct option)

☐ $4x - 3$

☐ $4x^2 + 1$

☒ $4x + 3$

☐ $4x^2 - 1$



correct

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Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min)

Let f and g be the functions defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$ then composition of f and g is

Answer (Please select your correct option)

☐ $6x + 6$

correct



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☐ $5x + 5$

☐ $6x + 7$

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Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min)

The negation of $1 < x < 10$ is $x \leq 1$ or $x \geq 10$ by using:

Answer (Please select your correct option)

☐ Distributive Law

☐ Inequality Law

☐ De-Morgan's Law

☐ None of these

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Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min)

If the n th term of a sequence is $a_n = 2(-3)^n + 5^n$ then the term a_1 is

Answer (Please select your correct option)

☐ -1

☐ 0

☐ 1

☐ 2

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Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

The part of definition which can be expressed in terms of smaller versions of itself is called

Answer (Please select your correct option)

☐ Recursion

correct



☐ Conclusion

☐ Base

☐ Restriction

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Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

The tower of Hanoi is a puzzle consisting of

Answer (Please select your correct option)

☐ 2 people

☐ 3 people

☐ 4 people

correct

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Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

The same element can never appear ----- in a set.

Answer (Please select your correct option)

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☐ twice

correct

☐ once

☐ thrice

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Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

If $(A \cup B) = A$ then

Answer (Please select your correct option)

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☐ $(A \cap B) = B^c$

☐ $(A \cap B) = A$

☐ $(A \cap B) = B$

correct

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Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)

If p = It is raining

q = She will go to college

"It is raining and she will not go to college" will be denoted by

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Answer (Please select your correct option)

☐ $p \wedge \neg q$

correct

☐ $p \wedge q$

☐ $\neg (p \wedge q)$

☐ $\neg p \wedge q$

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Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

Determine whether the function $f(x) = x + 1$ is one-to-one?

Answer ([Please click here to Add Answer](#))



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Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

Compute the first four terms of the sequence defined by the formula $a_n = 3n - 5$.

Answer (Please [click here](#) to Add Answer)



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Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

Show that $f(x) = x^3 + 1$ is onto function.

Answer ([Please click here to Add Answer](#))



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Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Find the sum of the infinite G.P $2, \sqrt{2}, 1, \dots$

Answer (Please [click here](#) to Add Answer)



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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Let $A = \{a, b, c, d\}$ be a set and consider the relation $R = \{(a, a), (a, b), (a, c), (a, d), (b, b), (b, d), (c, c), (c, d), (d, d)\}$ on A . Show that R is a partial ordering.

Answer ([Please click here to Add Answer](#))



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Let $f(x) = ax + b$ and $g(x) = cx + d$, where a, b, c and d are constants. Under what condition $f \circ g(x) = g \circ f(x)$.

Answer (Please [click here](#) to Add Answer)

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